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20 December 1971
SUPERSEDING
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(see 6.6)

MILITARY SPECIFICATION

DECODER GROUP, AN/UPA-59(V) IFF MK XII (SIF), WITH ACTIVE READOUT AND ASSOCIATED MODIFICATION TO INDICATORS (PPI)

1. SCOPE

1.1 This specification covers the construction and inspection of complete and operative lightweight decoder groups. Also covered under this specification are active readouts appropriate for the particular decoder group types and auxiliary functions such as automatic emergency alarm and selected altitude layer operation.

1.2 The decoder group shall accept IFF MK XII (SIF) video signals from the interrogator and decode the signals either passively or actively or both simultaneously. Passive decoding shall include both bracket and SIF codes. Active readout shall be accomplished on an SIF signal (including altitude signal) appearing within a range/azimuth gate on an associated indicator (PPI), and the active readout portion of the equipment shall display the SIF code numbers of the signal and also display the altitude decoded signal.

1.2.1 Enabling signals for active decoding will be initiated by a modification within the indicator (PPI), and the resulting area gate shall be displayed on this indicator (see 3.6.6.1.10).

1.2.2 Video signals supplied by the interrogator set shall, by plug-in arrangement, be accepted in tagged video (see 3.4.15.10) form or in conventionally separated form. This function shall be accomplished by including the necessary control circuitry for separated video and tagged video on a common signal circuit card. Selection of the mode of operation shall be accomplished by simple reversal of insertion of this card or other method as approved by the command or agency concerned. Identification markings shall be provided to show directly the selected mode of operation.

1.2.3 Provisions shall be included in the active decoder to accept and process a range azimuth readout gate from the Navy Tactical Data System (NTDS). Provisions to fulfill this requirement shall consist of inserting into the circuitry one or more plug-in cards supplied with the equipment and stored within the decoder unit. For gate characteristics (see 3.6.1.6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

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SPECIFICATIONS

FEDERAL

PPP-B-636	-Box, Fiberboard.
PPP-T-60	-Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	-Tape, Pressure-Sensitive Adhesive Paper, (for Carton Scaling).

MILITARY

MIL-C-17/6	-Cable, Radio Frequency, Coaxial, RG-11A (NWR-3S)
MIL-C-17/29	-Cable, Radio Freqeuncy, Coaxial, RG-59B/U(NWR-11)
MIL-P-116	-Preservation, Methods of
MIL-S-901	-Shock Tests, H.I. (HIGH-IMPACT), Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-D-3464	-Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-Q-9858	-Quality Program Requirements
MIL-E-16400	-Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-I-22960	-Indicators, Radar, Naval Ship and Shore, General Specification
MIL-C-39012	-Connector, Coaxial, Radio Frequency, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	-Definition of and Basic Requirements for Enclosure for Electric and Electronic Equipment
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-147	-Palletized and Containerized Unit Loads, 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-167	-Mechanical Vibrations of Shipboard Equipment
MIL-STD-415	-Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-461	-Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-470	-Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	-Maintainability Demonstration
MIL-STD-751	-Radar Outputs, Naval Ship and Shore Standard for

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MIL-STD-781 -Reliability Tests Exponential Distribution
MIL-STD-785 -Reliability Program for Systems and Equipment Development and Production
MIL-STD-1345 -Data, Measurement, in Support of Maintenance, Calibration and Repair of Electronic Equipments

HANDBOOKS

MILITARY

MIL-HDBK-217 -Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472 -Maintainability Prediction

PUBLICATIONS

MILITARY

NAVSHIPS 0969-125-0250-Electronic Equipment, Test, Preferred, General Purpose

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on data of invitation for bids or request for proposal shall apply.

FEDERAL AVIATION AGENCY

U.S. NATIONAL STANDARD FOR COMMON SYSTEMS COMPONENT CHARACTERISTICS FOR THE IFF MARK X (SIF) Air Traffic Control Radar Beacon Systems SIF/ATCRBS dates 27 December 1963.

(Applications for copies should be addressed to the Federal Aviation Agency, RD-75, Washington, D.C. 20360.)

DOD AIMS SPO (ASZIC)

DOD AIMS 64-900 Specifications for Mode 4 Input/Output Data. Technical Standard for the ATCRBS/IFF/MKXII System (AIMS).

(Applications for copies of DOD AIMS 64-900 and Technical Standard for ATCRBS/IFF/MKXII System (AIMS), should be addressed to DOD AIMS SPO (ASZIC) Wright Patterson Air Force Base, Dayton, Ohio.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago Ill. 60606).

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NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036).

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.5).

3.2 General requirements. The equipment shall be in accordance with the provisions of MIL-E-16400, except as hereinafter specified.

3.2.1 Description. The decoder group shall be provided with maintenance aids and publication materials as set forth herein and shall consist of the following primary functions as detailed herein:

(a) Decoder
(b) Set control
(c) Emergency alarm
(d) Selected altitude layer
(e) Active readout capability

The decoder group shall consist of the following separate items:

(1) Basic decoder, AN/UPA-59(V) (See 3.6 except 3.6.6.1.5(a))
(2) Emergency alarm speaker, BZ-173/UPA-59(V) (see 3.6.4)
(3) Active readout unit, ID-1447 (see 3.6.6 except 3.6.6.1.10)
(4) PPI modification kit, MK-1336/SPA (see 3.6.6.1.10)

When active readout units are not required operationally, the basic decoder shall be identical to that described in this specification and for operation with the active readout unit to avoid modifications other than the simple addition of the active readout unit (and appropriate PPI indicator modifications) should active type of operation be desired at a later time. The design of the basic decoder, the active readout unit and their interconnection, shall be patterned to permit both integral and remote operation of the active readout unit. To accomplish this flexibility, a parallel wired rear panel connector shall be supplied in the basic decoder for use with the active readout remote cable required herein (see 3.6.6.1.8). Source impedances for driving the active readout unit shall be designed to allow both integral and remote active readout operation simultaneously or individually. No circuitry or function shall be included in the equipment which will preclude the easy field addition of expansion to ten specially selected codes simultaneously. To facilitate such expansion the output connector for the active readout shall provide at least the following: (a) bracket video, (b) mode tags or gates, (c) reply trains in parallel form, that is, after decoding, (d) tagged video reset signal decode, and (e) challenged mode lines up to five modes which parallel the interrogator lines to the passive decoder, and (f) degarble output.

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3.2.1.1 The decoder group shall be fully compatible when operated in connection with interrogators and radar sets such as the AN/UPX-1, II and UPX-23. This equipment shall functionally replace Decoder Group AN/UPA-24, 24A, 24B, 24C and AN/UPA-50. In addition to providing the SIF decoding function, the equipment shall provide certain other decoding features as required herein. In addition to the conventional acceptance of individualized inputs of Modes 1, 2, 3/A, 4, and C, the decoder groups shall, by simple installation changes, be capable of accepting these modes in the form of tagged video. Mode 4 signals shall be processed to the extent specified herein.

3.2.1.2 Each basic decoder shall be furnished with a blank panel to replace the active readout unit when not supplied, or when the active readout unit is used remotely.

3.2.1.3 As required by the contract or order, the variation of the decoder group shall be as follows:

AN/UPA-59(V) 1 - Passive decoder (3.6, except 3.6.6.1.5(a))
AN/UPA-59(V) 2 - Active decoder consists of AN/UPA-59(V) 1 plus active readout (3.6.6.1.5(a))
AN/UPA-59(V) 3 - Active decoder consists of AN/UPA-59(V) 1 with remote readout (3.6.6.1.5(b))
AN/UPA-59(V) 4 - Active decoder consists of AN/UPA-59(V) 1 with both remote and integral readouts (3.6.6.1.5(c))

3.2.2 Figure 1 is for general information purposes in depicting the equipment employed in a typical installation with associated interrogator set, radar set, and video switchboard.

3.2.3 Part layout. The construction shall allow maximum accessibility to the internal parts and subassemblies. All parts requiring replacement at field level shall be constructed into, or a part of, a replaceable subassembly or module.

3.2.3.1 Maintenance concept. The maintenance concept falls into three (3) categories; organizational maintenance, field maintenance, and depot maintenance. Organizational maintenance is defined as that maintenance performed on equipment while installed and which is performed during periodic inspections. Organizational maintenance is limited to functional or operational checks to determine quality of performance and operational readiness; and periodic checks and repairs required to ensure to the maximum possible extent that the equipment will operate throughout the next periodic level. Field maintenance is defined as that maintenance required to isolate and repair faulty modules and subassemblies, adjustment and alignment of parts, and equipments. Depot maintenance is that maintenance required to completely disassemble, allow detail analysis, and repairs of equipment subassemblies and parts.

3.2.3.2 Equipment part layout. Test points shall be chosen to provide a straightforward, logical step-by-step troubleshooting sequence as well as a single end-to-end performance check. Maintenance considerations shall provide for the identification of where and how to break loop or feedback circuits to allow for proper signal injection with test equipment and a step-by-step

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troubleshooting, adjustment or alignment of the circuit. Test points shall be located so that all are accessible without moving or rotating the equipment on the bench. Those test points which provide end-to-end performance test on equipment installed, or completely assembled on the bench should be located on the front panel. Adequate spacing shall be provided between test points, and between test points and other parts, to eliminate the possibility of arcing, faulty readings of signals, or damage to equipment or parts. Test points shall be provided for essential waveforms and voltage measurements where terminals are not otherwise accessible. The test points shall be identified with a designation, for example, "TP-10". Provisions shall be made to accept all test probes with handles up to one inch in diameter and with a probe size of 0.08 inch ±0.002 inch, when equipment is in operable conditions.

3.2.3.2.1 The above requirement shall provide essential test points. However, signal inputs and outputs are an absolute minimum on all plug-in subassemblies. Test points for all direct current (dc) supply voltages shall be provided and shall be plainly labeled and conveniently located for ease of servicing.

3.2.3.3 Maintenance consideration. Maintenance consideration shall provide for the location of test points and the identification of procedures that will permit positive verification of equipment performance when installed. These considerations shall also provide for location of possible addition test points and for the identification of procedures that will permit positive verification and troubleshooting sequence of functions within the equipment, both singly and collectively, including end-to-end performance check of the assembled equipment. These maintenance considerations and procedures shall provide for a quality of repair, that will restore the equipment to an overall quality equal to or better than that required by the quality control requirements of this specification.

3.2.3.4 Fault isolation. The unit shall be provided with built-in fault isolation circuitry to enable in-position functional checking. The test circuits shall be as complete as practicable (consistent with the concept of light weight, simplicity) to allow go-no-go-testing.

3.3 Quantitative reliability requirement. The specified MTBF (θ_o as defined by MIL-STD-781) of this equipment shall be 1000 hours.

3.3.1 Reliability program plan. The contractor shall submit a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785. Particular attention should be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks, and time schedule. The proposed reliability program plan as approved and incorporated into the contract becomes the basis for contractual compliance. The plan, including reliability test plans, must describe an integrated effort within the total project; it shall provide information to show how the contractor will meet specified quantitative reliability requirements during development and manufacture. The reliability program plan shall be submitted for Government approval (see 6.3).

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3.3.2 Reliability prediction. The contractor shall submit a reliability prediction in accordance with the method Reliability Prediction Based on Stress Factors and Part Populations of MIL-HDBK-217 to the command or agency concerned. The prediction shall assume a 65°C temperature and a Class 4 environment. The contractor shall evaluate the prediction with respect to the reliability expected from a production unit. The processes and procedures required to assure production of reliable equipment shall be thoroughly documented (see 6.3).

3.3.3 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.3.4 Reliability reports. Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links shall be submitted to the command or agency concerned (see 6.3).

3.3.5 Maintainability program plan. The contractor shall submit a maintainability program plan in accordance with and in paragraph sequence with MIL-STD-470. The program plan shall be a separate and distinct entity and shall be submitted for Government approval to the command or agency concerned (see 6.3).

3.3.6 Maintainability prediction. A maintainability prediction shall be made in accordance with procedure II of MIL-HDBK-472 and submitted to the command or agency concerned for approval (see 6.3). An equipment repair time (ERT) of fifteen (15) minutes is required.

3.4 Design requirements.

3.4.1 To the maximum extent practicable with the size and weight requirements, the equipment shall be constructed so that the functions such as: line driver, coincidence and kill circuits, video processors, video mixers, and video amplifier-drivers, are each to maximum extent practical complete plug-in assemblies with test points at input and output of each plug-in. These plug-in assemblies shall provide for quick service by removal of the defective function. Each plug-in assembly shall be of the printed or etched wiring type which can be easily repaired. The design of the plug-in assemblies and plug-in cards shall be patterned so that no mechanical or electrical damage shall occur if they are improperly inserted. Maintenance aids as specified herein shall be supplied (see 3.4.28).

3.4.1.1 Nonstandard parts. Nonstandard parts shall conform to the requirements of MIL-E-16400 and the following (see 6.3):

(a) A list of all nonstandard parts used in the first article sample must be submitted for approval prior to delivery of the first article sample for Government evaluation.

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(b) Any subsequent nonstandard parts not included in the first article sample may, at the Government's option, require a complete reperformance of the first article inspection tests, or applicable portions thereof.

3.4.2 Microminiature construction. Microminiature parts and techniques shall be used to the greatest extent practical. All such circuitry must be approved by the command or agency concerned.

3.4.3 Environmental conditions. The equipment shall be capable of operation under the environmental conditions, Class 4 temperature range.

3.4.4 Electrical power characteristics. The equipment shall be designed and constructed to operate from a 115 volt (V), 60 hertz (Hz) input power source and from a 115 V, 400 Hz input power source (but not simultaneously) in accordance with the supply line voltage and frequency requirements of MIL-E-16400.

3.4.4.1 The alternating supply voltage may have a form factor within the range 1.25 to 1.02 and may have an amplitude factor within the range of 1.68 to 1.16. Form factor is defined as the ratio of the root-mean-square (RMS) voltage to the average voltage, and has a value of 1.11 for a true sine wave. Amplitude factor is defined as the ratio of the peak instantaneous voltage to the RMS voltage and has a value of 1.4.4 for a true sine wave. Line supply surges of 350 V persisting for 1 cycle shall cause no permanent damage to the equipment. Taps shall be included on the power transformers, for an adjustment at installation, to provide for operation at the line supply voltages of 105 V and 125 V.

3.4.4.1.1 The equipment shall meet the requirements of this specification with a line voltage regulation of ±10 percent about each voltage tap for voltages specified in 3.4.4. The equipment shall have a regulated power supply which shall provide for a continuous overload of at least 50 percent.

3.4.4.1.1.1 Primary overload protection shall be provided by fuses; and blown fuse indicators are required. Extractor post type fuse-holders, and glow lamp blown fuse indicators shall be provided. Provision shall also be made for over-voltage and cross-voltage protection of voltages used for integrated circuits.

3.4.4.1.1.2 Secondary power supply protection. The power supply outputs shall be self-protecting against faults which place an overload (above reserve capability) or short circuit on any or all of the power supply outputs. The power supply need not operate during the presence of such faults, but the use of fuses for this protection will not be allowed, and operation shall return to normal once the fault has cleared.

3.4.5 External connectors and controls. External connectors and controls in accordance with MIL-E-16400 shall apply. Connector covers shall be provided for each cable connector on the equipment. Video input and output connectors shall be type BNC in accordance with MIL-C-39012.

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3.4.5.1 The equipment shall be furnished with the following connectors and mating connectors:

(a) Mode 1 video
(b) Mode 2 video
(c) Mode 3/A video
(d) Mode C video
(e) Power
(f) Control (Mode 1, Mode 2, Mode 3/A select and challenge, and any other control wire as specified in 3.2.1.1)
(g) Control (Mode 4 control, Mode 4 override, Mode C control, 6.3V, and ground). Consideration shall be given to the use of a single cable fitted with dual connectors for the interrogator end to conserve rear panel space and satisfy the requirements for this control cable and the foregoing control cable.
(h) Emergency alarm speaker
(i) Mode 4 video
(j) Radar triggers

3.4.5.1.1 No cables are required. Mating connectors complete with inserts shall be supplied.

3.4.6 Power input requirement. The total power consumed, as supplied and measured at the power input terminals shall not exceed 75 watts when the equipment is operated under any combination of conditions in this specification. Additionally, the total RMS volt amperes as measured by sideband thermocouple meters shall not exceed 100 V amperes. This includes the power drain of the active readout which shall be at a minimum consistent with the performance requirements herein.

3.4.7 Cable entry. Cable entry shall be as follows: All connectors for power, control, and video shall be supplied and mounted on a removable connector plate(s) on the rear panel of the equipment.

3.4.8 Radio frequency (RF) shielding. In service the decoder will be operated near or interconnected with highpower radar equipment from which high frequency energy will be radiated and conducted to the decoder. Provisions shall be made in the equipment for proper operation under such conditions. Filtering shall be incorporated to assure proper operation under these conditions (see 4.3.7).

3.4.8.1 Electromagnetic interference emission, susceptibility, and equipment and systems selfcompatibility. The test set shall operate within the electromagnetic interference emission, susceptibility, and equipment and systems selfcompatibility limits in accordance with class IC of MIL-STD-461.

3.4.9 Shock and vibration. The equipment shall be capable of withstanding the shock requirements of Grade A, Class I, Type A, lightweight of MIL-S-901 and type I vibration requirements of MIL-STD-167.

3.4.9.1 Inclination. The equipment shall meet the inclination requirements of MIL-E-16400.

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3.4.10 Weights and dimensions. The weight of the equipment shall not exceed 30 pounds. The overall dimensions of the equipment, excluding fastening devices and screw heads, shall not be more than 6 (width) by 12 (height) by 18 (depth) inches including connectors. The 30 pound limit shall include the active readout function.

3.4.11 Degree of enclosure. The degree of enclosure shall be dripproof in accordance with MIL-STD-108.

3.4.12 External finish. All external surfaces shall be painted Navy enamel. This requirement shall not prevent the use of color coded markings to facilitate matching parts during installation, painted lettering for identification and instructions, nor shall this requirement necessitate over-painting of instruction plates or interconnecting cables subject to handling during assembly and disassembly of the equipment. All edgelited and operating front panels shall be painted or annodized black.

3.4.13 Mounting provisions including brackets and mounting plates. Proposed mounting provisions shall be submitted to command or agency concerned for approval as part of the panel layout and operating controls requirement of 3.4.19.

3.4.14 Controls and indicators.

3.4.14.1 All controls located on the front panel shall be classified as continually required. Those infrequently-used controls in the chassis shall be screwdriver adjusted.

3.4.14.2 Locking devices. Locking devices, where required shall be capable of retaining the controls in any given setting within the range of control. The locking and unlocking action shall be easily and quickly accomplished and shall not affect the setting of the control. When in the unlocked position, the locking device shall not interfere with the normal operation of the control. Where vernier controls are used, the locking device shall operate on both main and vernier controls if necessary to prevent damage.

3.4.14.3 An elapsed time meter is not required.

3.4.14.4 Instruction plates are required where necessary.

3.4.15 Code number assignments. The code number assignments (see 6.4.4) of the code wave form are as follows: the 12 pulse positions are divided into four groups of three positions each. Each group is assigned a letter and a subscript number designating the pulse position in the group. The positions in each group are spaced in increments of 2.9 microsecond (μsec), measured leading-edge to leading-edge at half voltage points. Each information pulse shall be within ±0.10 μsec of its referenced position. The pulse positions referenced from the leading-edge of the first framing pulse are as follows:

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Pulse	Position from First Framing Pulse in μsec
A1	2.9 ±0.10
A2	5.8 ±0.10
A4	8.7 ±0.10
B1	11.6 ±0.10
B2	14.5 ±0.10
B4	17.4 ±0.10
X	10.15 ±0.10
C1	1.45 ±0.10
C2	4.35 ±0.10
C4	7.25 ±0.10
D1	13.05 ±0.10
D2	15.95 ±0.10
D4	18.85 ±0.10

3.4.15.1 Mode 1. Mode 1 consists of framing pulses and five pulse positions composed of A_1, A_2, A_4, B_1, and B_2.

3.4.15.1.1 Mode 1 code number assignments. The appropriate code number is obtained by adding the subscript numbers of each group where an information pulse appears. The digits obtained are arranged in the group order AB to form the particular 2-digit code number. Code numbers 00 through 73 designate mode 1 codes.

3.4.15.1.2 Mode 2. Mode 2 consists of framing pulses and 12 pulse positions spaced in multiples of 1.45 μsec, referenced from the leading-edge of the first framing pulse. The seventh pulse position is not to be used. The number and position of information pulses present determines information content. Mode 2 replies may be received of six information pulses only. In such cases, the final two indications shall be 00.

3.4.15.2 Mode 2 code assignments. The appropriate code number is obtained by considering each group separately and adding the subscripts where an information pulse appears. The digits obtained are arranged in group order ABCD to form the particular 4-digit code number. When all pulses are absent from a group, the appropriate number is zero for that group. Code numbers 0000 through 7777 designate mode 2 codes. For example: Code 2435 contains one pulse in A group (A_2), one pulse B group (B_4), two pulses in C group (C_1 and C_2), and two pulses in D group (D_1 and D_4).

3.4.15.2.1 Mode 3/A. Mode 3/A consists of (a) framing pulses and six pulse positions composed of A_1, A_2, A_4, B_1, B_2, and B_4; or (b) framing pulses and 12 pulse positions composed of A_1, A_2, A_4, B_1, B_2, B_4, C_1, C_2, C_4, D_1, D_2, and D_4.

3.4.15.2.1.1 Mode 3/A code number assignments. The appropriate code number of (a) above is obtained by adding the subscript numbers of each group, where an information pulse appears. The digits obtained are to be arranged in the group AB to form the particular 2-digit code number. Code numbers 00 through 77 designate 3/A codes. For (b) above, group order ABCD forms code numbers 0000 through 7777.

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3.4.15.3 Emergency. Transponder controls may enable temporary modification of normal replies to indicate emergency conditions or for special identification functions. Characteristics of the emergency response shall be as follows:

Mode 1 Four pulse trains. The first train shall contain the Mode 1 code in use and the remaining three trains shall consist of framing pulses only. The framing pulses shall be spaced 24.64 ±0.10, 44.95 ±0.15, 49.30 ±0.20, 69.60 ±0.25, 73.95 ±0.30 and 94.25 ±0.35 microseconds from the first pulse.

Mode 2 Four pulse trains. The first train shall contain the Mode 2 code in use and the remaining three trains shall consist of framing pulses only. The framing pulses shall be spaced 24.65 ±0.10, 44.95 ±0.15, 49.30 ±0.20, 69.60 ±0.25, 73.95 ±0.30, and 94.25 ±0.35 microseconds from the first pulse.

Mode 3/A (a) Four pulse trains. The first train shall contain Code 77(00), regardless of the Mode 3/A code in use, and the remaining three trains shall consist of framing pulses only. The framing pulses shall be spaced 24.65 ±0.10, 44.95 ±0.15, 49.30 ±0.20, 69.60 ±0.25, 73.95 ±0.30, and 94.25 ±0.35 microseconds from the first pulse.

(b) Four normal Mode 3/A trains up to 7700 with spacing and tolerance as in (a) above

(c) Code 7700

3.4.15.4 Identification of position. The identification of position (I/P aircraft only) provides a single repetition of the reply code, when this is mode 1, at an interval of 4.35 μsec ±0.1 μsec from leading edge of the last framing pulse of the first train. This persists for a preset time period of 15 to 30 seconds and is repeatable at any time. It is done by a key on the transponder and sometimes using a microphone switch. For modes 2 and 3/A reply train consists of a single normal train followed by a single pulse 4.35 ±0.1 μsec from leading edge of the last framing pulse of the first train.

3.4.15.5 Mode 4 reply. A mode 4 reply consists of three pulses spaced 1.8 μsec ±0.2 apart, and each pulse will exhibit the characteristics of a single mode 4 interrogation pulse (see DOD AIMS 64-900).

3.4.15.6 Mode C reply. Mode C replies will have pulse characteristics and code configurations similar to mode 2 replies. Functionally, each individual code, when decoded, represents a discrete altitude of the interrogated aircraft; and this shall be displayed in feet (see US National Standard).

3.4.15.7 Special reply 7600. Special reply to interrogation from mode 3/A shall consist of code 7600. The intent of the message is to indicate failure of the aircraft communications equipment.

3.4.15.8 X-pulse. Special identification pulse may be present at 10.15 μsec position in mode 1, 2, and 3A trains.

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3.4.15.9 Time measurements with pulse as reference. Measurements of spacing between pulse and other time measurements, made with a pulse as a reference, shall be made with reference to the instant when the leading-edge of the pulse reached 50 percent maximum amplitude.

3.4.15.10 Tagged video. Tagged video is defined as a system which employs unseparated IFF video to which has been added, during receiver off-time, mode identification tags and reset signals. (Tagged video is expected to be expanded in the future to perform additional functions.) Interrogation pulses are used as an identification tag prior to zero range for each interrogation, and a characteristic reset signal occurs approximately coincident with the end of the receiver gate. The tagged video system operates as follows:

(a) The mode identification tag is decoded in the before (or degarbling) delay line and tag decoder. This resulting decode pulse shall:

(1) Allow the video for that receiver gate to be decoded or presented as video
(2) Activate that mode gate generator
(3) Lock out all other mode gate generators for the duration of this receiver gate
(4) Activate (through the mode gate generator) the proper decoder functions as required
(5) Activate readout as required

(b) During the receiver gate period, the decoder functions as directed by the setting of the controls.
(c) At the end of the receiver video gate, the reset signal is decoded in the before (or degarbling) delay line and tag decoder; and the decode pulse resets the operating gate generator and activates mode decoder for the next identification tag.

3.4.15.10.1 Identification tags. The identification tags shall consist of the following:

(a) Mode 1 - 2 pulses 0.8 ±0.1 μsec wide, spaced 3 μsec ±.25 μsec
(b) Mode 2 - 2 pulses 0.8 ±0.1 μsec wide, spaced 5 μsec ±.25 μsec
(c) Mode 3/A - 2 pulses 0.8 ±0.1 μsec wide, spaced 8 μsec ±.25 μsec
(d) Mode C - 2 pulses 0.8 ±0.1 μsec wide, spaced 21 μsec ±.25 μsec
(e) Mode 4 - 4 pulses 0.5 μsec ±0.1 wide, spaced 2 ±0.1 μsec (as referenced to first pulse), 4 ±0.1 μsec, and 6 ±0.1 μsec
(f) Side lobe suppression pulses may be present in all five modes. For modes 1, 2, 3/A, and C, these pulses will occur 2 μsec after the first pulse and be 0.8 ±0.1 μsec wide. For mode 4, the pulse will occur 8 μsec ±0.1 μsec after the first pulse and be 0.5 μsec ±0.1 μsec wide

3.4.15.10.2 Reset signal. The reset signal shall consist of three pulses. The first pulse shall be 8.0 μsec ±0.25 μsec wide and occur 2.0 to 3.0 μsec from the end of the receiver gate. The second pulse, 2.0 ±0.25 μsec wide, shall occur 10.0 ±.25 μsec from the half voltage point of the first pulse

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leading edge. The third pulse shall be 4.0 ±0.25 µsec and shall occur 15.0 ±.25 µsec from the half voltage point of the first pulse leading edge. The end of the receiver gate shall be defined as that point in time where the output video has decreased at least 24 dB from normal but no later than 3.0 microseconds after the video has decreased 24 dB. To meet the jamming requirements, the mode gate reset should be deactivated for a preset time prior to receipt of the reset signal. The jamming reset gate inhibit shall be adjustable from 600 to 4500 µsec by an internal control.

3.4.15.10.3 Amplitude. The identification tag and the reset signal amplitudes shall be essentially the same as the limited level of the video signals (see 3.6.1.6.1). Tags shall be within ±1 dB of video (IFF) signal available from the interrogator.

3.4.15.10.4 Identification tag decode tolerance. Mode 1, 2, 3/A, 4 and C tag decoders shall produce a tag decode when the proper pulse configuration is received over ±0.35 µsec of the ideal spacing. For ±0.65 µsec from ideal spacing, no decode shall be produced.

3.4.15.11 Side lobe suppression pulse. Presence of or absence of the side lobe suppression pulses shall in no way affect the performance of the decoder.

3.4.15.12 Reset signal decode tolerance. The reset signal shall produce a reset signal decode pulse when any pulse shall be ±0.35 µsec from its ideal position. For ±0.55 µsec from ideal position, no decode shall occur.

3.4.15.13 Inhibit action. Inhibit taps shall be provided on the (before degarbling) delay line and circuitry provided so that only proper signals will be decoded for identification tags and reset signals. All other signals shall be inhibited.

3.4.15.14 Video switching, modes 1, 2, 3/A and C. Provisions shall be made when any Mode 1, 2, 3/A, and C identification tag is decoded to cause subsequent SIF video to be processed in conformance with the mode identification tag. When the code-decode switch is in the code position, the SIF processing circuits shall be bypassed; and the SIF code video shall be inserted as an input to the video mixer. When the code-decode switch is in the decode position, the video following the appropriate identification tag shall be processed in conformance with the settings of the controls.

3.4.15.14.1 When Mode 4 is activated, the Mode 4 identification tag shall cause the following Mode 4 video to bypass the SIF processing function and be inserted as an input to the video mixer (see 3.6.1.6.3).

3.4.15.15 Provision shall be made for providing either the six mode challenge lines of 3.4.5.1 (f) and (g) or for a three wire output encoded to provide the six challenge outputs. The encoding shall consist of a diode in series with each output and the outputs of the diode pairs connected. Modes 1, 2, and 3 shall have the diode cathode fed to the decoder Modes 1, 2, and 3 outputs and Modes 4, C and 4 override shall have the diode anode connected to

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the decoder Modes 1, 2, and 3 outputs. Selection shall be solderless links to provide the encoded outputs when selected on Modes 1, 2, and 3 control lines.

3.4.16 Remote control. Relays shall not be used unless approved by the command or agency concerned.

3.4.17 Control circuit power. Power for all remote control relay mechanisms within the equipment shall be provided by a low-voltage regulated (within ±1 percent) dc source within the equipment for ±10 percent line voltage variations. There will be supplied 6.3 volts alternating current (ac) on control lines from the associated interrogator equipment whenever the interrogator is energized. This control signal shall be used to actuate the interrogator power on the indicator on the front panel. Although specified herein as 6.3 volts, the relay function shall operate on any interrogator supplied voltage of 3 to 6.9 volts, never exceeding 250 milliamperes (mA).

3.4.18 Controls, indicators, and panel layouts. A portion of the front panel, preferably the upper right hand area shall be reserved for inclusion of the active readout unit, and in addition the following facilities are required:

(a) Front panel facilities.

(1) Radar video ON-OFF - M^3IX switch (3 positions)
(2) Mode select switches for Modes 1, 2, 3/A, 4, and C (see 3.4.27)
(3) CODE, DECODE, OFF switch
(4) Mode 1 code, mode 2 code, and mode 3/A code. Code selector switches, one for each mode. Lamps shall be provided to illuminate SIF codes manually set by use of these switches
(5) Interrogator power on indicator (white)
(6) Challenge on, indicate lamps (blue) located near and for each associated challenge switch
(7) Panel illumination lamps in sufficient numbers to assure that all panel controls and panel lettering can be read with equal and sufficient illumination. Edge lighting and dimming facilities are required. Edge lighting may serve as decoder power on indication
(8) A continuously variable control, variable from 1 to 20 nautical miles, sued to increase and decrease the range (length) of the target sector gate
(9) Fuses (active and spares) with blown fuse indicators on active fuses
(10) I/P - OFF - X-pulse switch (toggle); a three-position switch to select X-pulse decode in conjunction with passive decode; and select I/P pulse decode as a separate function, or as an additional decode, with passive or bracket video (see 3.6.1)
(11) Bracket on-off switch; a two-position switch to select bracket decode output
(12) IF gain and long-short GTC controls are not required
(13) Two challenge switches, as follows:

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(a) Mode 4 override challenge; continuous sector (gated), off, and momentary as depressed.
(b) Active challenge; continuous, off, and momentary (same as sector challenge).

(14) Automatic emergency alarm indicator lights showing blue for 4-train and green for 7700 train
(15) One each pushbutton for (1) Military Emergency Test, (2) Civil Emergency Test, (3) Alarm Deactivation, and (4) 7600 test
(16) One each flashing amber indicating light for communication failure 7600 (Mode 3/A only)
(17) SAL Control and indication facilities
(18) Read Gate switch, ON-OFF

(b) Rear panel facilities.

(1) Control connector(s) (see 3.4.5.1 (g))
(2) Power input connector-with provisions for 115 ac power input, three wire synchro input and separate 115 Vac synchro reference
(3) ON-OFF-ON switch marked, "Power". In a position labeled, "Interrogator, decoder power shall be applied simultaneously with interrogator power. In a position marked, "Local," power shall be applied independently
(4) Radar video input connector (from radar)
(5) Tagged video input connector (from interrogator)
(6) Mode 1 input connector (from interrogator)
(7) Mode 2 input connector (from interrogator)
(8) Mode 3/A input connector (from interrogator)
(9) Mode C input connector
(10) IFF-RDR-MIX video out converter to (PPI)
(11) SIF/Altitude active readout output connector
(12) Emergency alarm connector furnishing audio and indicator lamps voltage
(13) Mode 4 output connector (to external evaluator). This must be terminated in 75-ohm load (see 3.6.1.6.3.2)
(14) Coaxial connector for active readout enabling gate input (for NTDS) or area gate trigger from modified indicator (PPI), depending on position of internal link quantity one. This connector shall be designated the Area/Active Trigger. Area gate sector (in degrees) control data to indicator (PPI) shall be included with the power input connector. Note: Control phase must agree with power phase
(15) Mode 4 processed video input, quantity one
(16) Spare video connector: quantity one, available but unwired; input to mixer amplifier
(17) Mode 4 input connector (from interrogator)
(18) Radar trigger input connector

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(19) Mode tags and reset input connector
(20) Selection switch marked "6 and 12 pulse", additionally labeled "Mode 2 or Mode 3 only" associated with the 12 pulse position

3.4.19 All internal chassis controls and adjustments shall be clearly labeled with function and component designation: for example, "R-128, 150Vdc adj."

3.4.20 Interchangeability. The interchangeability requirement of MIL-E-16400 shall also apply to subassemblies and plug-in assemblies. A minimum of circuit alignment is allowed when parts are replaced.

3.4.21 Specific parts interchangeability. It shall be possible to interchange transistors and crystals in a random manner with transistors and crystals meeting the requirements herein, without affecting the circuit performance in any significant respect.

3.4.22 Accessibility. The equipment shall be designed so that all parts are accessible for complete maintenance at the place of installation. The equipment may be of the drawer-type construction, and, if necessary, shall employ a pivoted type channel guide for withdrawal from the case and tilting to maintenance position. Provisions shall be made for chassis locking in the maintenance position.

3.4.23 Selectable delay. An adjustable method shall be provided internally as an installation selection to allow code video to be displayed with essentially no delay, or with delay of 20.4 µsec, or with delay of 40.7 µsec, or with delay of 61.0 µsec, all with tolerance of ±0.5 µsec. The output code pulse width shall be equal to the input pulse width with a tolerance of ±10 percent.

3.4.24 Reserve plug-in capability. The equipment design shall provide for 10 percent expansion. Space and cutouts with mounting facilities for additional plug-in cards shall be provided. A minimum of 2 spares is required. All printed card sizes and connectors supplied shall be identical. The service extension card shall be stored in one of these spaces.

3.4.25 Internal adjustments. Readily accessible internal adjustments for IFF video adjustment of 2 ±0.5 volts and other adjustments shall be provided to include, but not to be restricted to, the following:

(a) Passive decode, code and bracket decode level adjustment. (The output signal levels for the three signals shall be within 50 millivolts of one another.)
(b) Passive decode stretch adjustment
(c) Sector gate level adjustment
(d) Mixer amplifier output adjustment
(e) Passive degarble adjustment
(f) Readout hold duration adjustment
(g) Single/dual reply comparison link(s) adjustment
(h) A maintenance control provided to adjust the area gate sector gate between 1 and 15 degrees

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(i) Automatic emergency alarm adjustments
(j) Video input threshold adjustment
(k) I/P second pulse width (range) adjustment
(l) I/P spacing adjustment
(m) Internal Power Disable switch shall be provided for front access and located immediately behind the front panel

3.4.25.1 Appropriate internal test jacks shall be provided closely associated with the respective adjustments.

3.4.26 Video input termination connections and interconnections. Provisions shall be made in the decoder for employment of the equipment when used in parallel with other decoder groups and not as the last group in the system. Similar provisions and explanation shall be for employment as the final group, or the only group in the system. Emphasis is directed to the use of jumpers and terminating resistors. It is required that all interconnections between separately housed units involving video signals and triggers be at standard amplitude and impedance levels as specified in MIL-STD-751 and MIL-I-22960, where applicable. Inter-unit gates are also to be construed as subject to the video signal requirements of this standard.

3.4.27 Supplementary operational requirements. In addition to the functions described herein, multiple mode operation shall be available from the decoder as follows:

(a) Any one of the following modes or combination of modes shall be selectable; 1, 2, 3/A, 4 and C.
(b) Mode 4 Override, when challenged, shall override (a) above and shall provide a ground output on both the selected Mode 4 challenge line and the Mode 4 Override Challenge line. The sector gate shall be applied to the Mode 4 Select Line when the Mode 4 override switch is in sector position in order to insure multiple decoder operation.

3.4.28 Maintenance aids. Maintenance aids shall be supplied for each decoder group. Maintenance aids are defined as consisting of universal socket and adapters, cable adapters, and service extension cards. Contractor shall submit to the command or agency concerned, as soon as practical during first article design, a proposed list of these items (including proposed method of securing during storage) with brief, but essential, functional descriptions.

3.4.29 A self-test feature shall be provided by front panel selector switch(s) for the decoder group. Insertion, decoding, and display of the front panel selected test code configuration 0000 through 7777 on a one code at a time basis shall be considered adequate to test the passive, and active decode function and the display. This test feature shall also test the selected attitude layer and emergency alarm function, as described herein. Front panel selection shall also be provided for conversion of the inserted code to attitude in feet for the active readout display test of the Mode C conversion. This test switch shall be used for front panel test control and shall have three positions designated SIF, OFF and Mode C.

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3.4.30 Special tools. Special tools are not desired; however, any tools of a special nature, normally necessary for equipment assembly, disassembly, installation, or adjustment, shall be supplied. Provisions shall be made for the securing of these items, within the equipment with which they are to be used, to enable ready accessibility and minimize loss. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.4.31 Test equipment.

(a) Each equipment shall be provided with test fittings as necessary to permit maintenance, calibration, repair with test equipment selected for the purpose. General Purpose Test Equipment shall be selected from NAVSHIPS 0969-125-0250. Requests for other than the equipment specified in NAVSHIPS 0969-125-0250 shall be submitted in writing to the command or agency concerned for approval.

(b) The equipment shall be designed such that fault location and isolation of modules and boards shall be done through the use of General Purpose Test Equipment or a tester specified by the command or agency concerned.

3.4.32 Test points. Test points shall be provided in accordance with MIL-STD-415. Test points shall be provided to perform the following type of tests:

(a) Quality monitoring
(b) Fault location to align replaceable unit
(c) Fault isolation to a shop replaceable piece part
(d) System alignment, and
(e) Planned system maintenance test. Test point data shall be furnished in accordance with MIL-STD-1345.

3.5 Performance requirements. The decoder group, when operated with radar sets and interrogators specified herein, shall properly decode transponder replies received after processing by the interrogator. Demonstration of acceptable performance shall be the contractor's responsibility, and verification shall be through compliance with performance tests of 4.3.6. There shall be no change in range presentation of replies between equipment use as an active or as a passive device, or as simultaneously an active and passive device. When the active readout is de-energized, the range-strobe function of the indicator (PPI) shall be restored to normal and the area gate (window) not be visible on the indicator (PPI).

3.6 Function. The decoder group shall utilize, as a general guide for basic design, the functional block diagrams of figures 2 and 3 which are supplied for information only and are not represented as complete in all particulars of the requirements herein. Definitions used in this specification shall be as detailed in MIL-STD-751 (see 6.4). The decoder group shall process SIF coded replies, develop passive and bracket decode outputs, and display the active readout of reply codes. The active readout feature will serve for rapid code

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identification of an unknown IFF target. There shall be the capability to display simultaneously all SIF targets by means of bracket decode, and selected targets with passive decode. The passive-decode targets shall be stretched with respect to the bracket targets. When both selected target and brackets are desired, the selected target shall be stretched in pulse width for presentation. When either bracket or selected targets are used separately, normal decode pulse width shall be presented. The degarble circuitry shall reduce the probability of false target display or false readout to a minimum. No pulse regeneration or trigger circuitry shall be enabled until decision has been made by the decoder that a valid decode has been made. Mode 4 continuous will be challenged in readout gate azimuth only. Mode 4 selected by the Mode switch will be similar to other interlaced modes. In this mode of operation selected SIF codes would be stretched, Mode 4 and bracket decode, would not be stretched.

3.6.1 The decoder shall utilize delay lines and solid state techniques for processing multiple pulse-coded IFF trains. The decoder shall provide IFF video outputs, bracket, passive decode, and delayed or undelayed code (jumper selectable) for target display on an associated indicator (PPI). The decoder shall, in addition, decode up to 12 pulse codes in mode 2, and mode 3/A with interleaving and provide simultaneous bracket and passive decode output (passive stretched). I/P decoding and X-pulse decoding shall be provided as follows:

(a) If the I/P feature is selected, all reply trains (except mode C), having the I/P pulse, shall appear as a double slash target on the indicator (PPI). The I/P double-slash spacing presentation on the PPI shall be essentially the same for modes 2 and 3 as for mode 1, and the spacing variable by use of an internal screwdriver adjustment with limits of 4 to 40 μsec. The output pulse width for the second pulse shall be 0.45 μsec ±0.1 μsec or shall be screwdriver adjustable from less than 5 μsec to greater than 20 μsec when selected by a solderable link. When delivered to the government, the contractor shall set the position of the two 0.45 μsec pulses of the decode 1/P at 24.65 μsec.

(b) If the incoming reply code train has the "X" pulse position slot filled, the normal decode shall not be affected; however, if X-pulse parity is required by the front panel switch, a decode shall occur only if the X-pulse is present along with the correctly preset code.

3.6.1.1 Reply codes from interrogated targets shall be analyzed in the decoder by delay lines and semi-conductor logic. The circuitry, in addition to the main decoding delay line, must have before and after (degarbling) delay lines. Figure 2 depicts basic block diagram using tagged video. When tagged video is not in use, a plug-in module shall replace the tagged video circuits and allow use of conventional method of video and mode separation. One dual function plug-in unit may be supplied per decoder group to allow selection at time of installation for operation either using tagged video or conventional video. Circuitry shall be included in the tagged video plug-in to essentially eliminate the possibility of the tagged video reset signal being used as a jamming signal. Figure 2 also depicts the basic block diagram using before and after (degarbling) delay lines. It is required that the printed cards/delay

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lines (and associated circuitry) be arranged for each decoder in a design of one or more dual function plug-in unit(s) as necessary. This shall allow selection at the time of installation for operation using the before (degarbling) line or the normal decode line for passive decoding.

3.6.1.2 The incoming reply code trains shall be impressed across a 20.3 µsec long "before" delay line. The line shall be tapped at multiples of 1.45 with a tolerance of ±0.05 µsec from the nominal so that all pulse positions of a given train can be simultaneously examined. The output of the "before" line shall drive a second "decode" line, and the taps of this line shall be connected to semi-conductor AND and OR gates in the code select circuit of the passive decode function. These gates shall be biased to establish a specific logic matrix. Code select switch settings shall be used to set the logic matrix. This matrix represents the preset code, which requires the presence or absence of pulse as preset at each tap, for passive decode. If the input code train is equivalent to the preset code, a coincidence signal shall be generated and supplied to the passive decode circuitry. Beside this check for parity, a passive decode shall require bracket pulse coincidence. If the bracket start and stop pulses are detected across the decode delay lines, the bracket circuit shall generate a bracket coincidence read gate which shall be applied to the passive decode circuit. Simultaneous occurrence of a parity-coincidence signal, a bracket read gate, and absence of a kill signal shall satisfy the passive decode logic. A kill signal shall disable the passive decode and be derived from the code select circuits if the input reply code is not equivalent to the preset code.

3.6.1.3 The mixer amplifier section of the decoder shall combine the radar video, IFF video, and area gate video (when active readout is used). The output video from the mixer amplifier is then fed to the associated indicator (PPI). The output and input video may take front-panel-selected forms as follows:

(a) Undelayed or delayed IFF code video (installation adjustment) (see 3.4.23 and 3.4.26)
(b) Bracket video
(c) Passive decode video (1.0 µsec wide)
(d) Bracket plus passive decode stretched (15 µsec nominal, with internal adjustment of from 1 µsec to 20 µsec under all conditions of environment). The pulse width shall remain within ±25 percent of the first µsec of set value and ±15 percent for all additional stretch pulse width
(e) I/P decode video with double slash available for display
(f) Passive decode plus I/P
(g) Bracket decode plus I/P
(h) Bracket plus passive (stretched) plus I/P
(i) Radar video
(j) Radar video plus any one (1) IFF output of items (a) through (h)
(k) Mode 4 video and Mode 4 video link shall be provided to display either the Mode 4 video or Mode 4 processed video
(l) One spare video input capable of handling signals similar to radar video above
(m) Area gate video for indicator (PPI) as required and detailed herein

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3.6.1.4 A bypass shall be provided in the decoder to supply radar video directly to the associated indicator (PPI) if the decoder is deenergized for any reason. This shall occur independent of the position of the radar video on-off switch.

3.6.1.5 Degarbling. Electronic means shall be provided to recognize the presence of garble signals and suppress a readout when garble occurs. Readout of interleaved signal trains shall not be suppressed. The degarbling circuitry shall not inhibit the decoding and display of (a) I/P, (b) Emergency, and (c) mode C signals when the mode C signals include the SPI pulse. The degarbling feature shall be operative during active readout; however, a simplified technique such as card reversal shall be provided to allow enabling or disabling the degarbling during passive type of operation.

3.6.1.6 Video inputs. Provision shall be made for terminating each of the video inputs in 75 ohms ±5 percent of 1 watt. Each unterminated input impedance shall be not less than 3000 ohms and not more than 100 micromicrofarads for IFF and not more than 200 micromicrofarads for radar. Terminating resistors shall be as close as possible to the end of all video input cables, and their location shall be approved by the command or agency concerned. Terminating resistors shall be wired in or out by a wire jumper as an installation procedure.

3.6.1.6.1 IFF inputs. The decoder group shall operate with IFF input video signals, with characteristics as defined in MIL-STD-751 and the following:

(a) Positive polarity pulse trains
(b) Source impedance of a nominal 75 ohms ±5 percent
(c) Pulse trains of limited peak amplitude 2V ±0.5V (depending upon radar and IFF set employed). Signal or noise or both may vary from 0 to peak amplitude. To extract signals approaching the noise level, it is required that no active circuitry be employed to regenerate the signals prior to the decoding function.
(d) Positive detected signal plus noise of 2V ±0.5V (peak amplitude, see (c), with a duty cycle of signal plus noise of 0 to 25 percent). Signal may make up to 5 percent of total duty cycle.

NOTE: Duty cycle can be continuous, gated, made up entirely of noise, or entirely of signals, as well as a combination of any or all.

(e) The IFF pulses in the train may have the following characteristics:

(1) Duration: 0.30 to 1.0 µsec
(2) Rise time: 0.08 to 0.20 µsec
(3) Decay time: 0.08 to 0.30 µsec
(4) Pulse top: See MIL-STD-751
(5) Undershoot: Less than 5 percent
(6) Amplitude: Less than ±1 dB amplitude change between all pulses in any pulse train
(7) In general, pulse width within a single train shall remain constant (±0.1 µsec, or less).

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3.6.1.6.2 Radar inputs. The decoder group shall operate with radar input signals as defined in MIL-STD-751 and the following:

(a) From 0.30 to 1500 µsec duration
(b) The duty cycle of the signal plus noise, measured during signal time, shall be from 0 to 60 percent. The signal time may be continuous or gated on for a period from 20 to 5000 µsec at repetition rates from 10 to 5000 Hz.

3.6.1.6.3 IFF input selection. The decoder unit shall be capable of operating in the following manner:

(a) Accept mode 1, mode 2, mode 3/A, mode 4 and Mode C on separate input cables without mode identification tags
(b) Accept unseparated video plus mode identification tags and reset signal on one cable with amplitude as defined in 3.4.15.10.2
(c) Accept unseparated video on one cable with amplitude as defined in 3.4.15.10.2 and mode identification tags and reset signal on a separate cable. The amplitude of the mode identification tags and reset signal shall be between 2 and 50 volts positive and shall be adjustable, downward to meet the amplitude requirements of 3.4.15.10.3

3.6.1.6.3.1 The decoder shall accept the following additional inputs:

(a) Area gate range trigger (from PPI)
(b) Area gate azimuth data (from PPI)

3.6.1.6.3.2 Provisions shall be included to supply Mode 4 video (with characteristics compatible with 3.6.1.12 through 3.6.1.14) on the Mode 4 output connector, simultaneously with the Mode 4 input to the mixer amplifier.

3.6.1.6.4 Gating. Provision shall be made to generate gating signals to enable the active decode/readout as a function of manually selected range/azimuth information from the associated indicator (PPI).

3.6.1.6.4.1 Gate initiating signals will be supplied either in rho-theta or x-y form depending on the type of PPI used in association with the decoder equipment.

(a) NTDS Consoles will normally supply x-y gate signals -2 to -15 ±5V (75 ohm source) occurring with the x and y range dimensions of either ±2, ±4, or ±16 miles and centered on the point in space determined by the ball tab position. A link shall be provided for selecting the above negative input.
(b) All other PPI consoles will require field changes to provide gate enabling signals of 2 to 20 ±5V (75 ohm source) to occur in coincidence with the range strobe and cursor position each time the cursor is scanned. The gate enabling signal shall continue to be generated at the range strobe position for each sweep following the scan of the cursor position for a scan interval adjustment within the PPI field change.

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3.6.1.6.4.2 The equipment shall accept the signals of 3.6.1.6.4.1 (a) or (b) and generate a gate signal to enable the active decode function for an interval of 12 to 250 microseconds (1 to 20 nautical miles) as determined by the setting of the gate range control of 3.4.18.(a) 8.

3.6.1.6.4.3 Provisions shall also be included in the active decoder to accept and process a range azimuth readout gate provided from an external source. The gate will be on for that portion of the sweep range that is desired for active readout. The gate will be present on a sweep-to-sweep basis for only the azimuth sector that is required for active readout. The decoder gate connector shall exhibit a 75 ohm impedance, and the decoder shall function on a gate whose characteristics are -0.2 ±0.2V for true (on) and -4.8 ±0.2V for false (off).

3.6.1.6.5 Video input-AC coupling shall be incorporated in all video input circuits. DC restoration shall be utilized in the Radar, IFF Gating and spare video inputs to prevent baseline shift. The Radar/IFF Trigger shall not require DC restoration.

3.6.1.7 Passive decoding. A single nominal 0.45 µsec pulse output is produced for each pulse train of the proper configuration received. The reception of all other pulse-train configurations, on a one-at-a-time basis and under no noise conditions, shall not produce an output to exceed 30 dB below the proper output signal level; that is, the 2V level. The equipment shall be capable of decoding up to four pulse trains having 24.65 ±0.1 µsec between initial framing pulses. It shall be possible by manipulation of the bracket, decode, or code switches to cause the equipment to produce an appropriate output for each setting.

3.6.1.8 Individual pulse shift. When any nominal pulse (0.45 µsec width) of the correct configuration is shifted from its nominal position by ±0.2 µsec, the output from the decoder shall not decrease more than 10 percent from the nominal output of two volts at nominal spacing. For a pulse shift of ±0.5 µsec, the output of the decoder shall decrease more than 30 dB from the output at nominal spacing. Nominal spacing shall be considered ideal pulse position. When the pulse width is increased from 0.3 µsec to 1.0 µsec, the decode pulse amplitude shall be at least -6 dB from nominal output at less than a pulse width either side of nominal pulse position; and the slope of the coincidence curve shall be approximately parallel to the normal (0.45 µsec) curve.

3.6.1.9 Extraneous pulses. For extraneous pulses occurring within ±0.2 µsec of a space position in a correctly coded train, the output of the decoder shall be less than 30 dB below nominal output of two volts. For extraneous pulses occurring at ±0.5 µsec or more from a space position in a correctly coded pulse train, there shall be no measurable decrease in nominal decode level. Normal output is defined as the output of the unit with the extraneous pulses removed, and with ideal spacing for the correct pulse train.

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3.6.1.10 Interleaved pulse train. Any combination of proper pulse trains, of the nominal configuration whose pulses are interleaved so that any or all of the pulses may be spaced a minimum of 0.7 μsec between leading edges, shall each be decoded with less than -3 dB change in the decode amplitude from the normal output of the equipment with ideal spacing for a single correct pulse train. This shall not interfere with the stretched pulse condition. For interleave conditions the decoded output width shall be as specified in 3.6.3 with an additional 10 percent tolerance.

3.6.1.11 Decoding with noise. The reception of noise shall not generate an output from the decoder at a rate greater than one response per second on a plan position indicator. In the reception of a signal plus noise, with a duty cycle of 20 percent of which at least 5 percent is signal, there shall be no false decode or killing action as detected on a plan position indicator.

3.6.1.12 Mixer amplifier. The mixer amplifier shall accept the inputs as cited in 3.6.1.3.

3.6.1.12.1 The mixed output shall be adjustable from 1.5 volts to 2.5 volts.

3.6.1.12.2 Mixer amplifier gain. An internal adjustment shall be provided with a range of not less than 0.75 to 1.25 gain, but not greater than 0.5 to 1.5 gain. Unless hereinafter specified, all measurements shall be made with the gain of the mixer amplifier set at unity with a 2V signal and a duty cycle of signal plus noise of 30 percent.

3.6.1.13 The radar video signal shall be supplied directly to the output terminal, passing through no electronic circuitry when the switch is in the radar position, or when the decoder is de-energized. Radar signals with a duty cycle of signal plus noise as measured during signal time varied from 30 to 60 percent shall be passed as follows when in the mixed position.

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Input Signal	Allowable Amplitude Variations
0.25 to 0.5 volt	Plus 0, minus 3 dB
0.5 to 1.0 volt	Plus 0, minus 2 dB
1.0 to 3.0 volt	Plus or minus 1 dB

All signals of 30 percent duty cycle (of signal plus noise) or less shall have not more than ±1 dB amplitude change. There shall be less than 10 percent variation in rise time, duration, and decay time, and less than 5 percent overshoot; and no measurable undershoot shall be added, under all conditions of input signal variations with the selector switches in either the radar or mixed positions, or when the decoder is de-energized.

3.6.1.13.1 In both the mixed and IFF positions, the radar input shall be terminated in a nominal 75 ohms. By installation procedure, it shall be possible to remove this termination.

3.6.1.14 Mixer output. When terminated by a nominal 75-ohm impedance, the output of the decoder shall have the following characteristics:

(a) Positive polarity output
(b) Less than 5 percent change (for example: rise time, decay time, width, overshoot, and pulse droop) in coded IFF, decoded IFF, or radar signals, except as specified elsewhere
(c) Radar signals (see 3.6.1.13)
(d) Radar video frequency response shall be flat to within 1 dB (that is, 1 dB total, not ±1 dB) from 1000 Hz to 4 MHz, 3 dB (that is, 3 dB total, not ±3 dB) from 100 Hz to 10 MHz, and 6 dB (that is, 6 dB total, not ±6 dB) to 13 MHz. (Band width measurement shall be included in production testing.) The reference frequency for these measurements shall be 1 MHz/s
(e) Not more than 10 percent change in output level of any video signal, when IFF and radar signals are superimposed
(f) Cross-talk between input terminals or leakage within the equipment shall be at least 30 dB below peak-to-peak signal level as measured at the output for all positions of the radar, bracket, and code-decode switches. This is a peak-to-peak measurement during signal time, and includes both positive and negative amplitude developed by pulses with rise time and decay time equal to that of the mixer amplifier. Hum and spurious within the equipment shall be at least 30 dB below peak signal level as measured at the output for all positions of the radar, bracket and code-decode switches. This is a peak measurement during signal time, and includes either positive or negative amplitudes of all undesired spurious signals
(g) The decoder IFF output under noise conditions shall comply with 3.6.1.11

3.6.1.15 When sector challenge is used, targets located within the azimuth width of the sector gate shall be actively displayed. With the passive challenge switch on, continuous challenge shall be provided; and the IFF target display shall not be affected by the target sector gate. In either switch position,

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active readout of target replies is enabled only for time intervals equivalent to the size of the displayed gate. The active readout shall be retained on the channels until reset. A continuously variable internal adjustment shall be provided for active readout hold duration permitting reset time of a minimum of two seconds to a maximum of ten seconds following the active decode display.

3.6.1.16 The decoder, when operated with associated interrogator and radar, shall (in addition to accepting IFF video and supplying decode, bracket, code) supply sector-gate video outputs for display on the indicator (PPI). The target sector gate generated by the decoder will be used to select a specific target or targets to identify its unknown code. This sector gate shall enable readout of target reply codes on the channels of the readout. This sector shall also be displayed during the PPI video sweep and appear as brightened traces which define the periphery of essentially a rectangular shape with azimuth radials and range arcs. The area embraced by the sector gate shall be adjustable from less than one to more than twenty nautical miles in range and from less than one to more than fifteen degrees in azimuth. The position of the sector gate shall be selected by the cursor bearing control and by the range strobe control on the (indicator) PPI. The cursor and range counters shall indicate in degrees and nautical miles the bearing of the opening gate and near edge (range) of the sector gate on the PPI.

3.6.1.17 Composite mode trigger. The decoded Mode Tags or gates shall be combined to form a composite mode trigger for Modes 1, 2, 3 and C. The composite mode trigger shall be selectable by link strap to perform the functions of the radar trigger when the radar trigger is not available. The decoding and use of the composite mode trigger shall be compatible with the test generator and all other functions normally served by the trigger.

3.6.2 Spare gain. The equipment circuitry shall feature spare gain for operation with long cables. The design shall consider line compensation filters and provide satisfactory operation when the decoder is used with 1,000 feet separation from the interrogator. Such cables are not required to be furnished with equipment delivered; however, the technical manual shall set forth essential commentary on use with long cables and any special precautions, as necessary relative to routing of modes 1, 2, 3/A and C video, radar video, and control information. Such separation does not apply to the distance between the decoder proper and the active readout section. Spare gain considerations shall include use of RG-11B/U in accordance with MIL-C-17/6 and RG-59B/U in accordance with MIL-C-17/29 for coaxial cables.

3.6.3 Decode output. Provisions shall be incorporated such that the decode pulses may be fed after processing directly to the mixer amplifier; and, therefore, the width of the output pulse shall be equal to the width of the input pulse with a tolerance of ±25 percent. Provisions shall be made for 1 μsec ±25 percent pulse width at all times for any decode pulse, or continuously variable adjustable 1 to 20 μsec for each decode pulse.

3.6.4 Automatic emergency alarm function. An automatic emergency alarm function shall be supplied and an "enclosed speaker" suitable for mounting remotely to the decoder. The speaker assembly shall also duplicate the green and blue duplexed light indication function that is required for the decoder front

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panel. Activation of the 4-train test push button shall light the blue light and sound an appropriate tone for 8 seconds. Activation of the 7700 test push button shall be similar to the 4-train but with a green light. When both push buttons are activated, appropriate lights and tones shall be enabled. The deactivation momentary push button shall disable only audible alarms for a 2 to 10 second period, internally adjustable. After 10 seconds, even if the momentary push button is held, the alarm shall be reactivated. After recovery, the push button shall repeat this 10-second deactivation each time it is pushed. Upon receipt of a Mode 3/A Code 7600 reply, the alarm circuitry shall process the reply in a manner similar to that detailed herein for the emergency alarm signals, except the resulting voltage shall actuate the front panel 7600 flashing amber 8 second light. A front panel 7600 test push button shall be supplied. The speaker assembly shall also duplicate the amber light function. The alarm deactivation push button shall not affect the 7600 amber light indication.

3.6.4.1 Function. In conjunction with the prime decoder, this function shall be capable of performing the following functions:

(a) Resolve any 20.3 µsec pulse spacing, regardless of code combination, into a bracket decode
(b) Resolve any code 7700 into a decode output pulse
(c) Resolve four trains, any code, spaced 4.35 µsec (see US National Standards) into a single progressively decoded output pulse
(d) Provide an integration of (b) to a single pulse for 2 to 15 decodes within 0.2 second
(e) Provide an integration of (c) to a single pulse for 2 to 15 decodes within 0.2 second
(f) Provide an audible alarm when (d) occurs
(g) Provide an audible alarm when (e) occurs, distinct and different in tone from (f) above, such that a third distinct and different tone is generated when (f) and (g) occur combined
(h) Provide automatically a video signal for presentation on the indicator (PPI) when (b), (c), or (k), and (f), (g) or (m) or all occur
(i) Light the pilot lights of 3.4.18.(a)14. When (f) or (g) or both occur
(j) Provide a selective feature to require both (f) and (g) to occur before sounding the audible alarm (combined signal of (f) and (g)) and lighting both lights (i)
(k) Resolve any code 7600 into a decode output pulse
(l) Provide an integration of (k) into a single pulse for 2 to 15 decodes within 0.2 second
(m) Provide any audible alarm when (l) occurs which is the same tone as (f)

3.6.4.2 Detailed requirements. The IFF video inputs shall be those for the prime decoder; and the selection, such as challenge and turn on, shall be those as for the normal decoder. When tagged video is employed, the alarm shall function on all modes 1, 2, and 3/A which are challenged by the associated interrogator. When operating in the mode separated manner, only that mode

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called for by the particular decoder shall function (unless all mode bracket decode is used).

3.6.4.2.1 Decoding. Decoding shall be as follows:

(a) The basic function of the emergency alarm shall be to automatically present bracket decodes on the indicator (PPI) and sound an audible alarm when military IFF emergency SIF pulse trains are received
(b) A secondary function shall be to present a decoded code 7700 on the indicator (PPI) and sound an audible alarm when civilian aircraft emergency signals are received
(c) The military and civil signals shall be as specified in US National Standards and Military standards as specified herein
(d) Only the emergency signals specified in (c) shall activate the alarm

3.6.4.2.2 Individual pulse shift. Individual pulse shift requirements are as follows:

(a) For bracket decode, a single pulse output shall be generated for each pulse train of any code configuration which has brackets spaced 20.3 µsec. When the bracket pulse (of 0.45 µsec pulse width) is shifted ±0.3 µsec from this nominal of 20.3 µsec, the bracket decode output shall not decrease more than 10 percent of its value at nominal spacing. For a pulse shift of ±0.6 µsec from the nominal, the output shall be reduced at least 30 dB below the value at nominal spacing
(b) For a 4-train decode, the nominal spacing between the first pulse and the associated bracket pulse shall be 20.3 µsec ±0.05 µsec followed by three sets of framing pulses appearing at 24.65 ±0.1 µsec, 44.95 ±0.15 µsec, 49.30 ±0.20 µsec, 69.60 ±0.25 µsec, 73.95 ±0.30 µsec, and 94.25 ±0.35 µsec; all as measured to the first framing pulse. When a 4-train signal (of any code configuration) of the above spacing is decoded, the alarm shall be activated (as required herein). When the between-train spacing is nominal and for a pulse shift of any framing pulse as in (a), the alarm shall function on the bracket decode, except that the alarm shall always be activated for ±0.4 µsec and never for ±0.6 µsec at 25°C to 50°C. The decoding tolerances shall be such that the alarm is always activated for ±0.3 µsec, and never for ±0.65 µsec at other required temperatures
(c) For code 7700 outputs, the decode characteristics shall be the same (for any of the eight positions) as for bracket decode
(d) For both the 4-train alarm and the code 7700 alarm, as well as the decode pulses, when the pulse width is increased over 0.45 µsec (up to 1 µsec) or decreased to 0.3 µsec, the coincidence characteristics shall increase or decrease such that the decode pulse output shall be at least -6 dB from nominal output at less than pulse width each side of nominal position. The slope of the coincidence curve shall be approximately parallel to the normal (0.45 µsec) curve

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3.6.4.2.3 Interleaved pulse trains. Any combination of proper pulse trains, which are interleaved as closely as 0.7 µsec between leading edges, shall be decoded and operate this alarm with no measureable change in performance.

3.6.4.2.4 Decoding with noise. Decoding with noise shall be the same as required herein for the prime decoder. In addition, the alarm shall operate properly with a duty cycle of signal and noise of up to 10 percent.

3.6.4.2.5 Decoded output. A single pulse shall be delivered to the mixer amplifier for each bracket decode of a 20.3 µsec train automatically, regardless of control box settings, after a code 7700 or 4-train signal has been received, and shall be available for display each time it is received. These decode signals shall also be integrated for activation of the appropriate alarms. The width of this bracket pulse shall be 1 ±0.2 µsec. Code 7700 decoded output, a 4-pulse output, shall be delivered for each decode of code 7700. This output shall be available for mixing with the video of the decoder when the code 7700 alarm is activated. Width of this pulse shall be 1 ±0.2 µsec and spaced 12 µsec ±2 µsec. In addition a three pulse display shall be provided to the video mixer when the code 7600 alarm is activated. Width of this output shall be 1 ±0.2 µsec and the spacing shall be 12 µsec ±2 µsec.

3.6.4.2.6 Integrators. Four integrators shall be included in the emergency alarm function. Their function shall be as follows:

(a) The first integrator shall count the four train decodes and provide an integration adjustable from less than two decodes to greater than fifteen decodes (but less than 20) within 0.2 second
(b) The second integrator shall count the code 7700 decodes and provide an integration adjustable from less than two decodes to greater than 15 decodes (but not to exceed 20 decodes) within 0.2 second
(c) The third integrator shall, when the feature is selected, gate the decoded outputs of the 4-train decode and code 7700 decode, and count this resultant signal to provide an adjustable integration of 1 to 5 (but not more than 10) within 0.2 second
(d) The adjustments for the count of the four integrators shall be calibrated to within ±zero count for a count of 1, ±1 count for count of 2 through 7, and ±2 counts for count of 8 and over
(e) A free-running gating circuit shall be included in the emergency alarm to reset the three integrators. The period of this gating reset circuit shall be 0.2 second ±10 percent
(f) The fourth integrator shall count the code 7600 decodes and provide an integration adjustable from less than two decodes to greater than 15 decodes (but not to exceed 20 decodes within 0.2 seconds)

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3.6.4.2.6.1 For integrator control, the integrator output shall perform the following functions:

(a) The output from the first integrator shall perform several functions. These functions shall continue for 1 second +50 percent, -25 percent for each target. When the output from the integrator occurs each and every 0.2 μsec, then the function shall continue without interruption as follows:

(1) Sound the audible alarm
(2) Light a flashing panel light
(3) Provide a signal for the "AND" gate
(4) Activate bracket decode at the video output of the decoder (at the second decode)

(b) The output from the second integrator shall perform the same function for the 7700 decode as the first integrator, except that the decode shall be presented on the indicator (PPI) at the fifth decode or count set by the integrator, whichever is less.

(c) The third integrator shall perform several functions. These functions shall continue for 8 ±4 seconds. When the output is continuous from integrators 1 and 2, this output shall also be continuous. These functions are as follows:

(1) Sound a combined tone to the alarm
(2) Light both flashing lights

(d) The output from the fourth integrator shall perform the same function for the 7600 decode as the first integrator except that the decode shall be presented on the indicator (PPI) at the fifth decode.

3.6.4.2.7 Adjustments. The emergency alarm shall have the following adjustments which can easily be made with the cards inserted in normal position:

(a) Decode output which shall be adjustable from at least 1 to 3V. This output shall meet 3.6.1.3 video output requirements
(b) Integrate No. 1 count
(c) Integrate No. 2 count
(d) Integrate No. 3 count
(e) Disable adjustment, 2 through 10 seconds
(f) Audio gain adjustment shall be made by a common control on the speaker alarm unit. Each tone output shall be approximately the same level
(g) Emergency video inhibit adjustment 5 to 175 μsec

3.6.4.2.8 False alarm elimination. Because the effectiveness of the emergency alarm is reduced directly with false alarms, precautions and circuitry shall be included to reduce the possibility of false alarms to a minimum. It is also mandatory that all true emergency signals be recognized. No

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active circuitry: that is, such as blocking oscillators, multivibrators, and pulse regeneration, shall be used in the alarm decoding; as these circuits may cause the effectiveness of the alarm to be reduced.

3.6.4.2.8.1 In order to reduce the effectiveness of many false alarms, the following provisions and circuitry shall be provided:

(a) A gating circuitry shall be provided to by-pass the video input(s). The radar trigger shall be used to activate this circuit
(b) The radar trigger shall inhibit the video after each trigger for an adjustable time of 5 to 175 µsec (±5 percent). This provision shall be controlled by a conveniently located switch readily accessible on the rear panel. The adjustment may be made in two steps using an internal link

3.6.4.2.9 Speaker. There shall be supplied two watts of audio power into an 8-ohm LS-445/u speaker, or equivalent. The alarm function shall operate properly whether the speaker is connected or not. The audible alarm shall be gated with the .2 second flasher signal.

3.6.5 Selected altitude layers (SAL). There shall be provided appropriate circuitry and readily accessible operator's controls for passive (or active) decode method of operation in presentation on the indicator (PPI) of targets which fly within selectable altitude layers. The layers shall be selectable from -1,000 feet to 99,000 feet in multiples of 1,000 feet. In conjunction with the operator's selection of a specific altitude layer, the bracket decode switch "on" position shall allow bracket display on the indicator (PPI) of any and all targets within that layer. When the SAL switch is in the "OFF" position, all bracket decodes shall be presented on the indicator (PPI) regardless of altitude. In addition, with the SAL switch "ON", the code-decode switch in the "decode" position, and the bracket switch "ON", it shall be possible to present data combined with manually preset passive decodes of modes 1, 2 and 3/A stretched. The first pulse of the code video shall be the same range as a normal passive decode pulse. All conditions listed on figure 4 shall become part of this requirement.

3.6.5.1 (a) Altitudes of 99,000 feet and up shall be considered as one single layer.
(b) Altitudes of 0 to -1,000 feet shall be considered as one single layer.

3.6.6 Active decoder (SIF) altitude readout units. As required in the contract or order and detailed herein there shall be designed, developed and tested service test models of the following active readout units.

3.6.6.1 The functions of the active readout shall perform as delineated in the following paragraphs.

3.6.6.1.1 Active decoder. In its active decoding function, the equipment shall accept SIF video trains of 3.4.15.10 and 3.6.1.6.1. The characteristics of the IFF (SIF) signals shall be in accordance with the Technical Standard for the ATCRBS/IFF/MKXII System (AIMS). The readout unit shall provide the numerical

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display of the received IFF MARK X (SIF) signals. Circuitry directly associated with the active readout shall be located within the active readout unit with the exception of the recognition and comparison circuits, to obtain simplification of cabling. Logic circuitry shall be provided to compare at least two successive replies on the same mode from a given target before readout occurs, to minimize false readouts. There shall be supplied a link(s) adjustment to allow, in one position, comparison for active readout on two successive replies, and in the other position, active readout on single replies without comparison. The readout unit shall be capable of receiving rho-theta video inputs of x-y video inputs. It is desirable that the foregoing be accomplished on a single printed circuit (PC) board or circuit.

3.6.6.1.2 SIF/Altitude readout unit. This readout unit shall provide circuitry to accomplish the following at the operator's option:

(a) Numerical display of SIF code video replies for modes 1, 2, 3/A, and altitude reporting. Only those replies shall be displayed which fall within an area gate selected by the operator at the radar indicator. Numerical display shall consist of 4 vertically-stacked direct-reading horizontal channels located on the front panel of the readout unit. Each channel shall contain five digits. The first digit shall be used for MODE designator, the remaining four digits shall be used as code-designator. All replies shall be preceded by the dynamic visual tag (see 3.4.15.10)

(1) On single target multiple mode interrogation, the channels shall be arranged modes 1, 2, 3/A and C in descending order

(2) Multiple targets single mode. More than one target within the selected area gate shall be read with a single selected mode. The readout shall be arranged with the first reply appearing in the top channel, second reply appearing in the second channel, third reply appearing in the third channel and the fourth reply in the fourth channel

(3) Dual target, dual mode with targets range correlated and range ordered; and presented as such

(b) Numerical display of Mode C. Mode C replies will indicate (in decimal feet) aircraft altitude to the nearest 100 feet. Of the four digits, the first three shall indicate thousands of feet, and the last indicate hundreds of feet. The last digit shall appear with a decimal preceding the display. This decimal marking shall be illuminated only for Mode C operation and shall be inconspicuous at all other times. The upper limit for altitude indication shall be 126,700 feet and the lower limit shall be minus 1000 feet. An appropriate "negative" indication shall appear for altitude below sea level

(c) Each channel of readouts shall contain mode indication capability for 1, 2, 3/A and C

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3.6.6.1.3 The readout units shall be supplied with one readily accessible control to allow operator selection of the brightness level of all digits within the channels. Consideration shall be given to the use of electronically switchable devices.

3.6.6.1.4 The display on the readout unit shall be as selected by the operator. For example, when front panel mode selector is set at modes 1, 2, 3/A and C only those modes selected shall be displayed.

3.6.6.1.5 The active readout unit shall be mechanically configured as a part of the decoder front panel as detailed in 3.4.18. This placement of the active unit shall be identified as the normal mode of operation. However, the electro-mechanical design shall be arranged such that with the use of the 10 foot active readout remote cable, as required herein, and the output connector (decoder rear panel), the following modes of operation shall be available:

(a) Normal mode as defined above
(b) Basic decoder operation with the active readout unit operable remotely up to 10 feet away
(c) (a) and (b) above combined when a second active readout unit is employed

3.6.6.1.6 The readout unit electrical power characteristics and power drain shall be as set forth in 3.4.4 and 3.4.6.

3.6.6.1.7 The readout unit weights and dimensions shall be as set forth in 3.4.10.

3.6.6.1.8 The readout units shall be packaged by type as cited herein for the prime decoder, and 10 foot cables as follows (with connectors wired) shall be supplied: coaxial cable for area gate triggers, control cable for area gate sector, and active readout remote cable.

3.6.6.1.9 A brief but essential description of the proposed physical and electrical characteristics of the readout functions shall be submitted to the command or agency concerned for approval prior to fabrication.

3.6.6.1.10 When required by the contract or order, the scope of work herein shall include modification kits for indicators (PPI) associated with the decoder group for active readout operation. First article samples of modification kits for all indicated indicators shall be submitted for approval. Production modification kits shall be supplied as required by the contract or order. Complete production drawings shall be submitted for modification kits for each indicator listed herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the equipment being procured. The quality assurance system shall be in accordance with MIL-Q-9858. A quality program plan, as it would apply to the requirements of this contract, shall be prepared prior to first article inspection. If first article inspection is not required, the quality program plan shall be prepared prior to production inspection. This program plan shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the quality control system or the quality program plan shall be submitted to the cognizant Government Inspector for approval prior to use (see 6.3).

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspection. The methods of examination and testing are as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)
(3) Environmental tests (see 4.4.3)

(c) Reliability testing (see 4.2.2)
(d) Inspection of preparation for delivery (see 4.5)

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4.2.1 Test procedures. The contractor shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and quality conformance inspection to the command or agency concerned, via the Government inspector for review and approval. Each test shall be identified in accordance with 4.2 as to classification (see 6.3).

4.2.1.1 Procedures for inspections, tests, and demonstrations, shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.2.1.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.2.1.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2.2 Reliability demonstration. Reliability demonstration shall be performed in accordance with MIL-STD-781.

4.2.2.1 Reliability demonstration test plans. The contractor shall submit to the command or agency concerned for approval a reliability demonstration test plan which conforms to the detailed requirements of section 5 of MIL-STD-781, and the requirements of this specification (see 6.3).

4.2.2.2 Reliability qualification tests. Qualification testing shall be performed on at least two (2) production units. Testing shall be performed at test level C with test plan IV. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan IV.

4.2.2.3 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to ensure continued production of acceptable equipment. Sampling testing shall be at test level C, with test plan IVA of MIL-STD-781. At least three (3) units shall be randomly selected from each four month's production for reliability sampling tests. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan IVA.

4.2.2.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.2.2.5 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.2.2.6 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the

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production tests of section 4 to assure that it meets the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

4.2.2.7 Reliability test procedures. The reliability tests shall be conducted using the procedures, schedules and performance limits specified herein as a minimum.

4.2.2.8 Burn-in. The contractor may elect to operate the reliability test samples for a certain time prior to the initiation of reliability testing. Any such burn-in, applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered nonrelevant but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the test samples.

4.2.2.9 Maintainability demonstration. Maintainability demonstration shall be performed in accordance with Method 3 of MIL-STD-471.

4.2.3 Verification of maintenance manual test procedures.

(a) The test procedures of the maintenance manual for DEPOT, intermediate and operator levels of maintenance shall be verified. The test equipment used to verify the procedures of the manual shall be the same as specified in the manual. Test equipment not specified in the manual may be used for the manual verification tests if the contractor makes a written request and the command or agency concerned approves the request.

(b) The manual verification shall proceed as follows:

(1) A technician shall give a complete set of boards, modules and components all of which are completely unaligned.
(2) The technician will align each module and determine that each module is operating properly. These tests will be done with the boards not installed in the prime equipment.
(3) The technician will install the components into the system and align the system.
(4) The technician will verify all the procedures of the maintenance standards book and technical manual.
(5) The technician will operate the system in accordance with the operators manual and determine that the system can meet the operating parameters.

(c) Manual verification:

(1) shall be performed by an electronics technician (ET) selected by the Navy and trained by the contractor.
(2) shall be performed before the units go into production.

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4.3 First article inspection. Unless otherwise specified (see 6.2) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection tests shall be in accordance with the listed first article tests and inspection procedures of MIL-E-16400, except as specified herein. The following tests shall not be performed:

(a) Water cooling
(b) Accelerated life test
(c) Noise test
(d) Nuclear air blast
(e) Reliability

The first article inspection tests may be equitably shared and scheduled among the first article samples to reduce time and cost elements, wherever possible, subject to approval of the proposed schedule by the command or agency concerned.

4.3.1 Operating test. The operating test shall be in accordance with MIL-E-16400, to determine conformance with this specification. Duration of the electrical operating test shall be 24 consecutive hours without interruption or failure.

4.3.2 Enclosure test. Enclosure test shall be conducted in accordance with MIL-E-16400 to demonstrate compliance with 3.4.11. The enclosure test shall be performed after the shock and vibration tests have been conducted.

4.3.3 Temperature test. The temperature test shall be Class 4 of MIL-E-16400. Verification shall be made during the temperature test, of compliance with 3.4.4.1, regarding line supply surges, line regulation, and 50 percent overload factor.

4.3.4 Humidity. Humidity tests shall be conducted in accordance with MIL-E-16400. Additional temperature and humidity tests shall be made, in accordance with MIL-E-16400, when deemed necessary by the Government.

4.3.4.1 Performance parameters to be measured shall be those equipment parameters approved by the command or agency concerned from a proposed list submitted by the contractor concurrently with the test procedures requirement of 4.2.1.

4.3.5 Interchangeability. The equipment shall be tested to determine conformance with the performance requirements specified herein after a complete change of subassemblies, or random but limited interchange of various transistors and crystals (see 3.4.21).

4.3.6 Performance tests. Performance tests shall be made by the contractors, in addition to the first article inspection, to determine equipment conformance with the performance requirements, as specified herein. These tests shall include, but not be restricted to, the following: Laboratory tests to demonstrate proper operation in decoding modes 1, 2, 3/A, and C (both as tagged and as conventional IFF video); and means of mixing the radar video, if desired,

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by the operator. Signals shall also be shown in the coded or decoded form for any mode. Simulated mode 4 and mode C signals shall be employed. Emergency signals for both Military and FAA shall be shown, and also special replies such as I/P and bracket decoding.

4.3.6.1 When active readout is procured, demonstration of operation of this section and the modified indicators (PPI) shall evidence capability in allowing an operator to select and read out actively the reply codes of modes 1, 2, 3/A and C as detailed in 3.6.6.

4.3.6.2 Noise test. In order to verify the equipment decoding and display capabilities (see 3.6.1.11), appropriate tests shall be performed using the AN/UPM-101 or equivalent as the noise source.

4.3.7 In order to verify the RF shielding required by 3.4.8 the Government tests conducted on the first article samples will include operation in the immediate vicinity of high power radar sets.

4.3.8 Design acceptance. If the results of the first article inspection and performance tests are acceptable to the command or agency concerned such acceptance shall constitute conditional approval of the design for purpose of shipment for Government evaluation. Final design approval for production may be granted, after satisfactory completion of the Government evaluation. Such acceptance shall not relieve the contractor from the responsibility of meeting the requirements specified herein. In the event tests by the contractor or the Government reveal deviations with respect to the requirements as specified herein, changes as may be required for the equipment to conform to the requirements of this specification shall be incorporated by the contractor into the equipment. Retests may be required, dependent on the nature of such modifications, at the discretion of the command or agency concerned.

4.3.8.1 The first article sample(s) shall be completely overhauled and refurbished at the completion of all first article inspection tests, and prior to shipment of the samples to the Government for service evaluation. In addition to the above requirement and in the event that the first article samples are required by the contract or order to be resubmitted as production equipment, the contractor shall thoroughly recondition the equipment before resubmitting it as production equipment, following the Government evaluation.

4.3.9 First article inspection report. A first article inspection report shall be prepared containing all the results of the first article inspection. The report shall include completed Form NAVEXOS 3373 as specified in MIL-S-901 for the shock test and resilient mount data.

4.3.9.1 The first article inspection report shall contain the following information:

(a) A technical discussion relating to each examination and test with a summary of these results. Compliance or noncompliance with each requirement of equipment specification shall be so stated

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(b) A block diagram of each test setup, including a description of the type of test equipment employed and dial settings
(c) A statement of the accuracy of the measurement taken under each test phase
(d) Supplementary data to include all data recorded in the performance of these tests, preceded by a summary table of this data
(e) A detailed list of all corrections to the equipment design necessitated during the testing phase, to comply with specification requirements
(f) Summary preface sheet. A summary preface sheet shall be provided stating all areas (with cross reference to related detailed tests) of noncompliance with requirements. All areas shall be interpreted to include each paragraph of sections 3 and 4 of this specification.
(g) The first article inspection report shall be signed by at least the test engineer, project engineer, and the Government inspector prior to submission.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.2.1 and shall be performed in accordance with MIL-E-16400. The duration of the operating test shall be eight consecutive hours without interruption or failures on each complete system, prior to delivery of that system. A condensed report of these results for each equipment shall be prepared and submitted to the command or agency concerned.

4.4.2 Production control inspection. Production control inspection shall be in accordance with MIL-E-16400 as modified to meet the applicable requirements for first article inspection, as specified herein. Sampling procedures for production control inspection shall be in accordance with MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection utilizing an AQL of 6.5 percent. Production control inspection shall be performed on equipments that have passed production inspection and shall be performed in accordance with test procedures approved in accordance with 4.2.1.

4.4.2.1 If at any time a sample unit fails in any PCI test the contractor shall be required to perform a failure analysis and make corrections as necessary to meet the requirements of this specification. The equipment shall then be retested. It is not required that additional equipments be subjected to PIC tests if there are failures within the PCI sample lot (this does not prohibit the contractor from testing additional equipment at his expense as part of a failure analysis).

4.4.2.2 If the failure analysis and corrective action discussed in 4.4.2.1 results in a change of part(s), circuit layout or the mechanical equivalent such

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action shall be reported to the command or agency concerned immediately. It shall then be determined by the command or agency concerned whether the correction should be made to the equipments already delivered under the contract. No production deliveries shall be made until the initial sample lot has successfully completed the PCI tests unless otherwise specified by the command or agency concerned. For subsequent lots production deliveries shall not be withheld pending completion of PCI tests or pending receipt of Government decision on retroactive corrections unless otherwise stipulated by the command or agency concerned.

4.4.3 Environmental tests. Environmental tests in accordance with MIL-E-16400 are required in the production cycle and shall be performed on equipments which have been subjected to and passed production inspection. Exception - accelerated life test is not required. The conditions for withholding production deliveries outlined in 4.4.2.2 for the production control inspection tests also apply for the environmental tests.

4.4.3.1 Environmental test samples. Samples for the environmental inspection shall be randomly selected in accordance with the following:

(a) One sample from the first 25 production units fabricated
(b) One sample from each successive 75 units produced, or fraction thereof

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the test specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes and so forth. If the government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor can not determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed production control and environmental tests may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with section 5. First article packs which contain Level A packages shall be subjected to rough handling (see 5.1.1).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.1.3 Preservative application. None.

5.1.1.4 Unit packaging. Each decoder group shall be individually packaged in accordance with Method II b of MIL-P-116, ensuring compliance with the General paragraph under Methods of Preservation (Unit Protection) and the Physical Protection paragraph therein and meet the rough handling tests of MIL-P-116. Cushioning and dunnage shall be used to prevent physical or mechanical damage. Desiccant shall conform to MIL-D-3464, Type II or III. The container exterior shall conform to PPP-B-636. Technical manuals shall be packaged in a waterproof, transparent, heat sealed bag, minimum 4-mil thick, and shall not be placed within the barrier used to package any other items.

5.1.2 Level C. Each complete decoder group shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Decoder groups packaged as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, V Grade, Special Use Requirement, for Type 2 Load. In lieu of the closure and waterproofing requirements in the appendix to PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, corners, and manufacturer's joint with waterproof tape, 2 inches minimum width, conforming to PPP-T-60, Class 1, or PPP-T-76. Reinforcing shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. Decoder groups packaged as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, Class weather-resistant. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. Items, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Unitized loads. When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated

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in shipping containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.2). Marking shall include special marking for Method II packs.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Intended use. This specification is intended for use in the construction and inspection of complete and operative lightweight decoder groups.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of first article samples to be submitted if other than specified in 4.3.
(c) Levels of preservation-packaging and packing, and applicable marking (see section 5).

6.3 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Technical manuals including operating instructions are to be provided with each equipment if required in the contract or order.
(b) Reliability program plan (see 3.3.1).
(c) Reliability prediction report (see 3.3.2).
(d) Reliability reports (see 3.3.4).
(e) Maintainability program plan (see 3.3.5).
(f) Maintainability prediction report (see 3.3.6).
(g) Nonstandard parts report (see 3.4.1.1).
(h) Quality program plan (see 4.1.1).
(i) Test procedure reports (see 4.2.1).
(j) Reliability demonstration test plans (see 4.2.2.1).

6.4 Definitions.

6.4.1 Definitions. The following definitions concern the general characteristics of information employed or processed by the equipment of a specific radar identification (IFF) system. The subject system, by means of a

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pulse-coded radio frequency transmission from one equipment (the interrogator), elicits a pulse-coded identification reply from a second equipment (the transponder). This information, when processed by the interrogator, is then decoded by a third equipment (the decoder).

6.4.2 Interrogation. An interrogation is two or more properly constituted radio frequency pulses transmitted by an interrogator set.

6.4.3 Reply. A reply is a radio frequency pulse or pulse group transmitted by a transponder set as a result of an interrogation.

6.4.3.1 Reply transmission characteristics. The characteristics of the reply transmission of modes 1, 2, 3/A, and C identification of position (I/P), and emergency replies are defined as follows:

6.4.3.1.1 Basic reply pulse. The basic reply pulse of modes 1, 2, 3/A, and C, is as follows:

(a)	Pulse width	0.45 ± 0.1 μsec
(b)	Rise time	0.05 to 0.1 μsec
(c)	Decay time	0.05 to 0.2 μsec

6.4.3.1.2 Framing pulses. Framing pulses (also referred to as brackets) are two basic pulses spaced 20.3 ± 0.1 μsec. Start (ST) designates the first framing pulse. Stop (SP) designates the last framing pulse.

6.4.3.1.3 Information pulse. A basic pulse contained between framing pulses and located with ± 0.10 μsec of its referenced position.

6.4.3.1.4 Pulse position. A time-restricted location of information pulses referenced from the first framing pulse, measured from leading-edge to leading-edge of half voltage points of pulses concerned. Each position is designated by a letter with a subscript number.

6.4.3.1.5 Pulse train. Pulse train is a group of basic pulses in specified pulse positions resulting from one interrogation or response.

6.4.4 Code. A code is a combination of information pulses designated by a mode and code number (see 3.4.15).

6.4.4.1 Pulse characteristics. The standard video and trigger pulse characteristics are as defined in MIL-STD-751.

6.4.5 Delay. Delay is the time interval measured between the 50 percent peak amplitude points of the leading edges of the pulse concerned.

6.4.6 Interleaved. Two or more trains of pulses whose time positions are such that the pulses of one train occur during the interval between pulses of the other train(s).

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6.4.7 Garble. Garble is defined as a form of pulse train signal interference caused by the overlapping (excluding interleaving) of two or more pulse trains, or the presence of one or more pulses in any of the nominal pulse position intervals during, preceding, or following a legitimate signal train by one train length.

6.4.8 Extraneous pulses. The phrase, "extraneous pulse," is used to designate pulse of approximately the same characteristics as the standard reply pulse except not appearing at proper position for standard trains.

6.4.9 Dynamic visual tag (DVT). An automatic readout of mode designation preceding the numerical display of active readout.

6.4.10 Progressive decoding. Progressive decoding of IFF trains is defined as storage of bracket decodes to be used as a gate or gates for successive pulses or trains. Gate width is determined by intertrain or pulse spacing tolerance. This provides minimum decode zones for brackets and varying intertrain spacings.

6.5 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to these bidders offering a product which has been previously procured or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.6 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETION, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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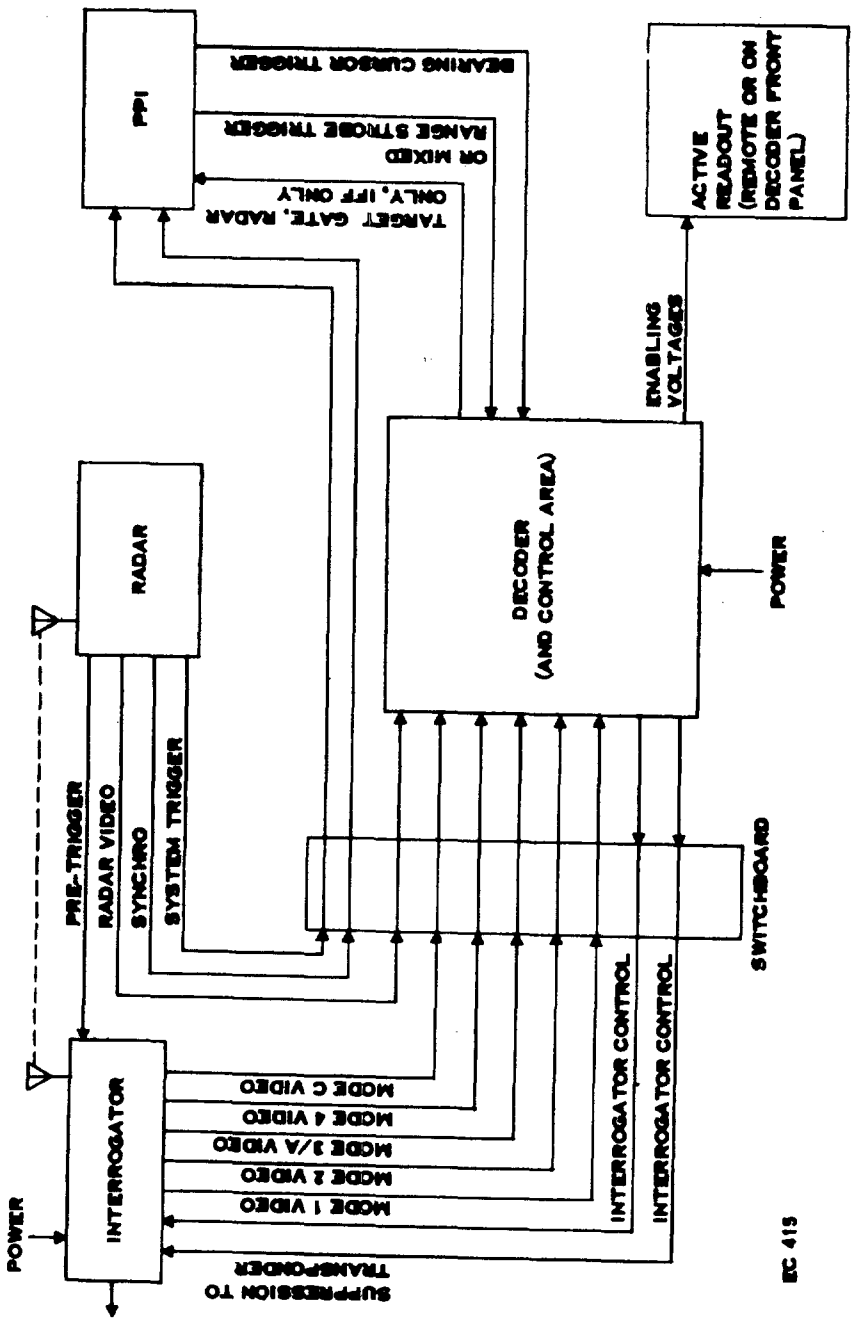


FIGURE 1. Typical decoder installation (not using tagged video)

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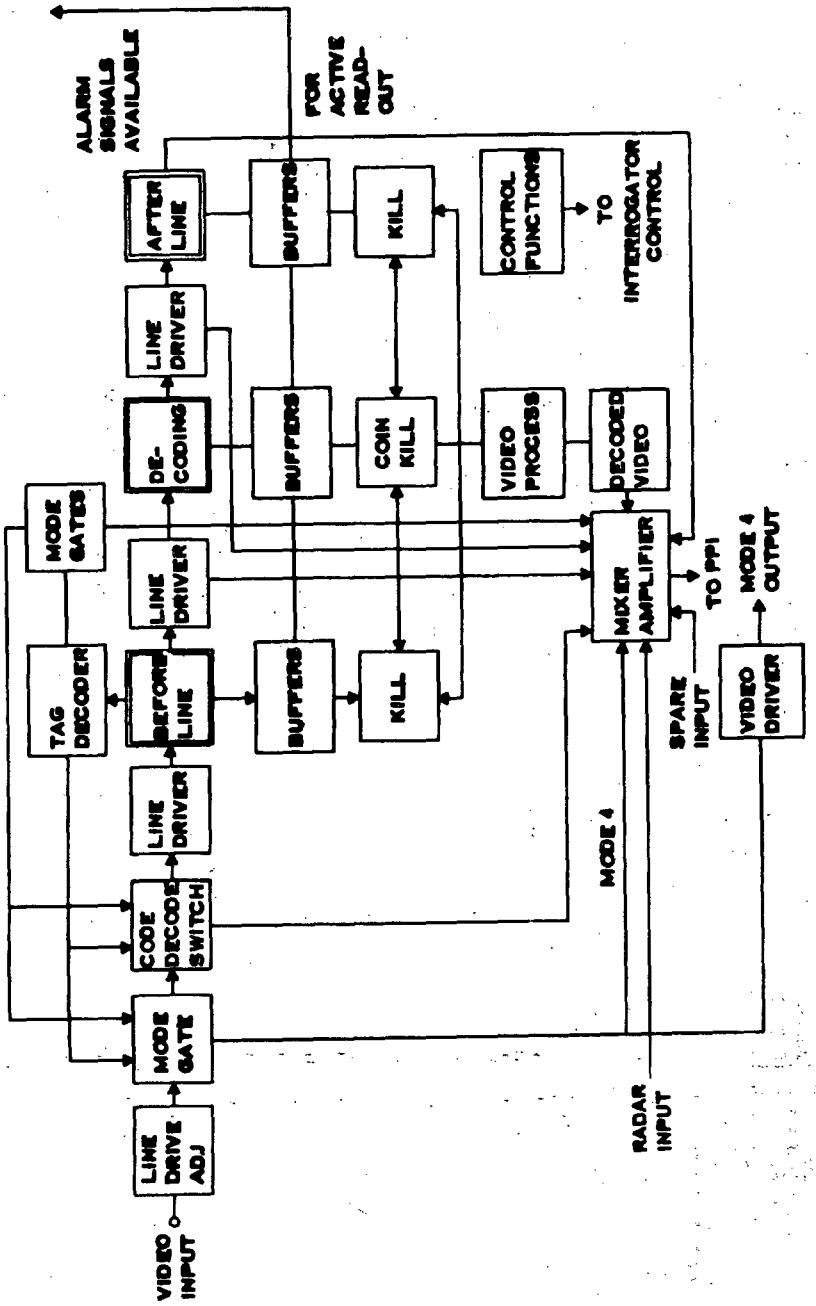


FIGURE 2. Functional block diagram of decoder (employs tagged video, degarbling used and allows fucure expansion)

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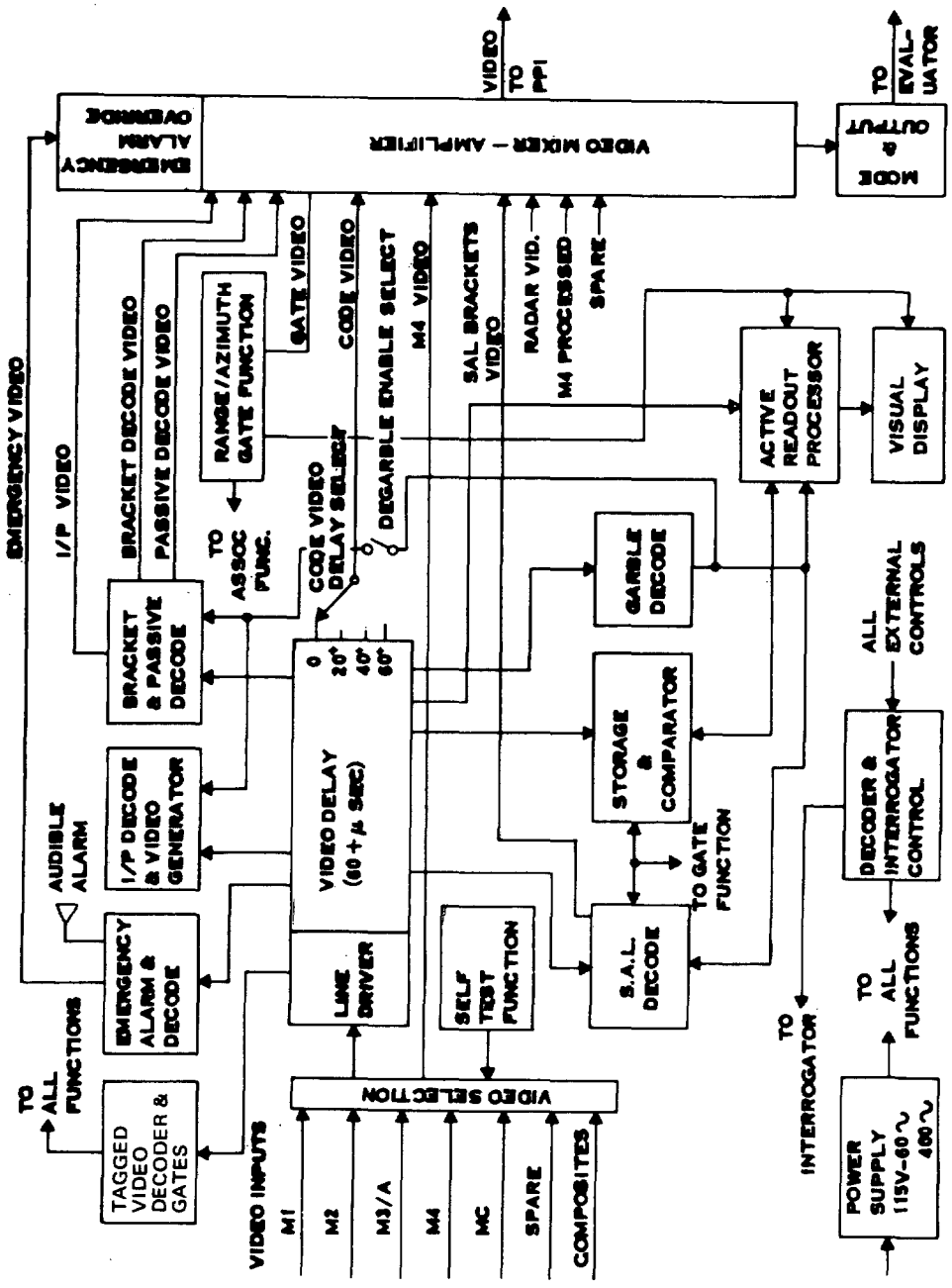


FIGURE 3. Simplified block diagram of IFF decoder

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FIGURE 4. Operator's control functions

	SAL	STRTCH	BRKT	DECODE OFF CODE	DISPLAYED VIDEO
1	ON	ON	ON	DECODE	All Mode C within layers - all passive decodes stretched All other targets standard bracket
2	OFF	ON	ON	DECODE	All passive decodes stretched no SAL All other targets standard bracket
3	ON	OFF	ON	DECODE	All targets - no differentiation for SAL or passive decodes
4	OFF	OFF	ON	DECODE	All targets - no SAL and no difference for passive decodes
5	ON	ON	OFF	DECODE	All passive modes and all SAL stretched
6	OFF	ON	OFF	DECODE	All passive modes stretched
7	ON	OFF	OFF	DECODE	All passive modes and all SAL standard arc
8	OFF	OFF	OFF	DECODE	All passive modes standard
9	ON	ON	ON	OFF	All targets standard arc
10	OFF	ON	ON	OFF	All targets standard arc
11	ON	OFF	ON	OFF	All targets standard arc
12	OFF	OFF	ON	OFF	All targets standard arc
13	ON	ON	OFF	OFF	All SAL stretch
14	OFF	ON	OFF	OFF	NOTHING
15	ON	OFF	OFF	OFF	All SAL standard arc
16	OFF	OFF	OFF	OFF	NOTHING
17	ON	ON	ON	CODE	Raw Video
18	OFF	ON	ON	CODE	Raw Video
19	ON	OFF	ON	CODE	Raw Video
20	OFF	OFF	ON	CODE	Raw Video
21	ON	ON	OFF	CODE	Raw Video

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FIGURE 4. Operator's control functions (continued)

	SAL	STRTCH	BRKT	DECODE OFF CODE	DISPLAYED VIDEO	
22	OFF	ON	OFF	CODE	Raw Video	
23	ON	OFF	OFF	CODE	Raw Video	
24	OFF	OFF	OFF	CODE	Raw Video	

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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